

To get started, let us initially set up users and passwords to configure security in Grub2. Open the `/etc/grub.d/00_header` and add the following commands to set up security based on authentication:

```
cat << EOF
set superusers="osfy"
password osfy l1nux
EOF
```

The aforementioned statement creates a super-user with the name 'osfy' and assigns the password 'l1nux' to it.



**Note:** It is not necessary to have the same user name present in the current Linux system. You can have any ambiguous name to secure the Grub system, irrespective of its availability as a system user.

To set up multiple users, you can simply assign multiple users and passwords with the password field:

```
cat << EOF
set superusers="osfy"
password osfy l1nux
password linux ubUntU
EOF
```

The password mentioned above is in plain text and can be read by anyone with administrator privileges. For further security enhancements and fortifications, Grub allows encryption of the password using an inbuilt tool, namely, `grub-mkpasswd_pbkdf2`. Running the command will bring up a prompt asking you to enter the password. Once entered, it will generate an encrypted string. To use the generated string, replace it with the password in front of the user of choice in the `/etc/grub.d/00_header`:

```
cat << EOF
set superusers="osfy"
password osfy grub.pbkdf2.sha512.10000.9FF12B1401811430D664B6
2473FE2CE19FA6B04563FE908ED7CF0462 E60CB06BD94320662B8C4D0
16609A4F613B2C45BEAE2C0BB823944D4C8A205C3EA78E093.9047CD06FE8
52574B9AEFBACD1275700557066A0A7EE307C98765E04E6997056D748BF21
C369617B94BB2DD3C096FA541A5F4C19BDD0F4D731D3785A16409652
password linux ubUntU
EOF
```

The string generated is too long and looks messy. Perhaps developers should fix the shortcomings of generating such long strings. Once you have got the user setting fixed, define which user should be able to boot or which entries should be secured.

To configure Grub2 further, you will have to edit `/etc/`



Figure 4: A customised Grub2 splash

`grub.d/40_custom`:

```
# (1) OSFY_Linux
menuentry "Open Source For You" --users osfy
set root=(hd0,2)
linux /vmlinuz-linux root=/dev/sda2 ro nomodeset
initrd /initramfs-linux26.img
```

Simply append `--users <user_name>` in front of `menuentry` and that segment will be secured. Make sure you enter a valid username; else it can result in boot failure.

**Customising Grub2:** The next step in managing Grub2 is to customise it. Unfortunately, Grub2 is not as customisable as the legacy Grub. Intuitive and fancy configuration options like Gfx Grub are missing at the time of writing, but might reappear in the near future. Grub2 offers some nifty cosmetic changes out-of-the-box, such as changing the Grub menu background and changing its colours.

To change the Grub background, copy a `jpeg`, `png` or `tga` format image in the `/boot/grub` folder. Once you have saved the file, open the configuration file `/etc/defaults/grub` with a text editor and add the following statement:

```
GRUB_BACKGROUND="/boot/grub/test.png"
```

Replace `test.png` with the filename that you have saved. Append the changes to the main configuration file by updating Grub. Use one of the following commands as the super user:

```
update-grub
grub-mkconfig -o /boot/grub/grub.cfg
```

To change the menu colours, add the following command:

```
GRUB_COLOR_NORMAL="red/white"
GRUB_COLOR_HIGHLIGHT="black"
```

To check the supported colour and gradients, please visit GNU Grub Theme Site: [http://www.gnu.org/software/grub/manual/html\\_node/Theme-file-format.html#Theme-file-format](http://www.gnu.org/software/grub/manual/html_node/Theme-file-format.html#Theme-file-format)